ALEIX CASAS BARCO

Full Stack Software Engineer

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SUMMARY

Full Stack Software Engineer with 3 years building product end to end on a SaaS urban analytics platform: from what the user needs and the architecture of each feature to its implementation, testing and release, working with AI agents at every step. I designed the architecture that lets clients embed any part of the product in their own websites and led an AI diagnosis feature on Gemini from proposal to production. A recent MSc in Software Development and Architecture gave me a solid base in software design and architecture.

SKILLS

- **Architecture:** DDD, event sourcing, modular monoliths, microservices, event-driven architecture, SOLID and design patterns, scalable system design
- **AI-assisted development:** AI coding agents, spec-driven development (spec, technical plan, implementation, tests)
- **AI in product:** LLM integration in production (Google Gemini), context design, cost controls (quotas, spend limits, caching), safe rendering of model output
- **Languages:** Python, TypeScript, JavaScript, SQL
- **Backend:** Django, Django REST Framework, Celery, Node.js, REST API design, JWT authentication, asynchronous processing
- **Frontend:** Angular 20 (signals, RxJS, standalone components), React, HTML, CSS
- **Databases:** PostgreSQL, Redis, BigQuery, data modeling, query optimization, indexing, partitioning
- **Cloud and Infrastructure:** Google Cloud Platform (GKE, Cloud SQL, BigQuery), Kubernetes (HPA, KEDA), Terraform, Docker, Nginx, GitHub Actions, GitLab CI, CI/CD
- **Observability and Testing:** Prometheus, Grafana, Sentry, load testing, pytest, Vitest, Playwright, unit, integration and end-to-end testing
- **Ways of working:** Agile, Scrum, Git, Jira, code review, mentoring

EXPERIENCE

Full Stack Software Engineer · Idencity, Barcelona | June 2023 - Present

Product team of 3 engineers building IDDIGO, a SaaS urban analytics platform with over 1,000 city and territory indicators. Each of us owns an area: I take features from the user need and the architecture to production, using AI agents throughout.

- **Embeddable widget platform.** Designed the architecture that lets any feature of the Angular application be configured, shared and embedded in client websites as a widget. A single entry point loads each widget's configuration, route guards keep visitors inside the functionality that widget allows, and a dedicated JWT authentication with proactive token refresh runs alongside the platform's existing authentication. Embedding is limited to each client's allowed domains through a Content Security Policy enforced at the Nginx layer. Extended it with territory pages configurable per widget and custom typography per client. Live in production.
- **AI diagnosis with Google Gemini.** Led the project from proposal to production as project owner and delivered the MVP in one month: a tab on every indicator page that writes an analysis of trends, anomalies and territorial comparisons. Built the structured context sent to the model, asynchronous generation with Celery, cost controls (a global daily spend limit and per-institution quotas), a 30-day cache tied to the data version, versioned reports with user ratings and token-usage tracking, and safe rendering of the model's output in the app and as a PDF.
- **Angular 20 migration and frontend testing.** Migrated the frontend to Angular 20 (standalone components, new control flow, functional guards and interceptors) in a 650-file change that removed over 5,000 lines net, moved application state to signals, and wrote the team's conventions for reactive code. Rebuilt frontend testing from scratch with Vitest and Playwright, now used by the whole team.
- Built **Django REST APIs over PostgreSQL holding tens of millions of records** and cut the heaviest endpoints from several seconds to under one through query and index optimization. Added Redis caching with automatic invalidation after every data import.

- Unified the platform's corporate PDF generation into a shared template base used by all four document generators, and built full data downloads for observatories and the data catalogue.
- Mentored two junior developers through code review, pairing and technical growth plans.

MASTER'S THESIS PROJECT

Vista Flota · Real-time IoT fleet management platform on Google Cloud | La Salle URL, 2026
Vista Flota lets fleet operators follow their trucks in real time. An IoT device on each vehicle publishes telemetry over MQTT (position, speed and fuel), which the platform ingests, deduplicates and turns into live fleet state, trips and alerts on an operator dashboard, while archiving the full history for route reconstruction and analysis.

Team project across three areas (Application, Infrastructure, IoT). Owner of the Application area.

- Built the event processor and data layer of a real-time ingestion system and took it **from 80 to 5,000 events per second at peak and 3,200 sustained with 0% message loss**, against a requirement of 1,000 events per second.
- Diagnosed and removed bottlenecks through a measure, diagnose, fix and re-validate cycle instrumented with Prometheus and Grafana: moved deduplication out of PostgreSQL into Redis to clear the critical path, added PgBouncer multiplexing 200 client connections over 20 real ones, implemented watermark-based backpressure to eliminate out-of-memory crashes, and switched MQTT to QoS 1 to stop message loss on pod restarts.
- Replaced KEDA autoscaling with Kubernetes HPA after profiling proved the processor pods were CPU-bound.
- Designed polyglot persistence: Redis for hot fleet state under 1 ms, Cloud SQL for relational data, and date-partitioned BigQuery for historical telemetry.
- Built the Angular operator dashboard and the backend REST API.
- **Stack:** TypeScript, Angular, Kubernetes (GKE), Terraform, MQTT/EMQX, Cloud SQL, BigQuery, Redis, Prometheus, Grafana, GitHub Actions.

EDUCATION

MSc in Software Development and Architecture · La Salle, Universitat Ramon Llull | Oct 2025 - Jul 2026
Software architecture, microservices, event sourcing, scalable system design, Kubernetes, NoSQL, testing, agile methodologies, AWS.

BSc in Computer Engineering · Universitat Autonoma de Barcelona | 2020 - 2024

LANGUAGES

Catalan (native) | Spanish (native) | English (B2, professional working proficiency)